Light Distillation for Calibration-Permitted Low-Rank Attention Compression

An optional companion to GRC, with falsifiable predictions

William Ken Ohara Stewart

NagusameCS Independent Research

nagusamecs@gmail.com https://github.com/NagusameCS/HyperTensor

April 2026 (v0.2, scaffold)

Abstract

Paper A introduced GRC, a calibration-free shared-basis low-rank projection of attention weights that is faster than the uncompressed model on a consumer GPU at k=1024 (+6.27% decode throughput, 95% bootstrap CI [1.0607, 1.0650], $p < 10^{-10}$, n=8 paired runs) on Llama-3.1-8B-Instruct Q4_K_M, but pays a measured +13.30% perplexity penalty at the throughput-preserving rank k=1536 (absolute PPL 6.79 → 7.69 on WikiText-2 [1]). This paper specifies an optional, opt-in distillation step that aims to recover a configurable fraction of that penalty in exchange for a few hundred forward passes on a small calibration corpus. The contribution is a protocol, not new mathematics: per-layer rank-r LoRA adapters [2] fit on the teacher–student logit residual after GRC projection, with sink-channel exemption [3, 4] as an orthogonal lever. We describe the design, derive a first-order bound on the achievable PPL gap closure (§ 3), prove that the runtime fusion path of Paper A is preserved under any of three documented merge strategies (§ 4), and specify the falsifiable predictions (§ 5) that an EC2 GPU run will confirm or refute. Status. The Phase 1 reference (calibration-free, CPU-only) is operational; the Phase 2 PyTorch runner (calibration-permitted, A100/L40S) is a scaffold. Empirical numbers in § 6 are predictions, not measurements; they will be replaced once the runner has executed. The paper is published in this scaffolded form so the protocol can be reviewed before the run, not after.

Contents

1 Introduction **3**

1.1 Project nomenclature . . . 3

1.2 Contributions . . . 3

1.3 What this paper is not . . . 4

2 Method **4**

2.1 Phase 1: GRC projection (unchanged from Paper A) . . . 4

2.2 Phase 2: Teacher–student LoRA correction (new) . . . 4

2.3 Phase 3: Merge and ship . . . 5

3 First-order bound on PPL gap closure **5**

4 Why distillation does not break the fusion path **6**

5 Falsifiable predictions **6**

6 Empirical protocol **7**

6.1 Done . . . 7

6.2 Pending: Phase 2 PyTorch runner . . . 7

6.3 Pending: Empirical numbers . . . 7

6.4 Pending: Throughput re-measurement . . . 8

7 Threat model and limitations **8**

7.1 What the Phase 2 run will not establish . . . 8

7.2 Implementation hazards . . . 8

8 Reproducibility **8**

1 Introduction

Calibration-free compression is a useful design point because it eliminates the data, the optimisation budget, and the per-deployment configuration that PTQ schemes such as GPTQ [5], AWQ [6], SmoothQuant [7], ASVD [8], SliceGPT [9], FWSVD [10], and SparseGPT [11] all require. Paper A shows that this design point is reachable for the $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V$ attention projections via PCA of the joint Gram matrix, at a measurable quality cost. The cost has two components, identified in Paper A's Limitations: an intrinsic spectral floor set by the Gram spectrum itself (the bottom 9% of joint Q+K+V energy on Llama-3.1-8B), and a reducible component attributable to the calibration-free choice of basis weighting and to the absence of any post-projection refinement.

This paper exercises the largest reducible lever, a small amount of post-projection distillation against a frozen teacher, behind a single opt-in flag. The design constraint is that turning distillation on must not break Paper A's runtime path; turning it off must reproduce Paper A's numbers exactly. The constraint is satisfied by construction because the distilled correction is a per-layer LoRA factor of rank $r \ll k$, mergeable into the projected weights and re-quantisable to Q4_K_M without the runtime ever observing a code change.

1.1 Project nomenclature

This paper inherits the four-level naming hierarchy of Paper A: HyperTensor (umbrella), Geodesic Projection (overarching pipeline), Geodesic Runtime Compression (GRC, the attention-only calibration-free subset, the subject of Paper A), and OTT/GTC (the longer-term manifold-runtime framework, Paper D). The present work is a strict extension of GRC: it adds a single Phase 2 between Paper A's projection and Paper A's runtime, leaves both endpoints unchanged, and is not in the critical path of any Paper A claim.

1.2 Contributions

1. A teacher–student protocol that augments calibration-free GRC with a parameter-efficient correction, framed as a $3 \cdot r \cdot d \cdot L$-parameter LoRA fit on the projected weights against a frozen teacher.

2. A first-order bound (§ 3) showing that the achievable PPL-gap closure is bounded above by the residual energy that GRC discards, evaluated on the calibration corpus, and below by zero. The bound is constructive: it gives a target number to compare the empirical gap closure against, so a Phase 2 run that closes substantially less than the bound is evidence that the LoRA capacity, the optimiser schedule, or the corpus size is the bottleneck rather than the projection loss itself.

3. Three documented merge strategies (§ 4) that preserve the GRC fusion path under distillation: (a) add-and-revert (no fusion, debugging only), (b) fused-base plus side-LoRA (recommended default; $O(r/k)$ overhead), and (c) re-quantised SVD-folded merge (lowest VRAM, highest merge-loss).

4. Three falsifiable predictions (§ 5) that the empirical run will adjudicate, written so that any one of them failing changes the conclusion of the paper.

5. An EC2 launch script and a CPU-only reference, both already checked into the repository. The CPU-only reference reproduces Paper A's GRC numbers exactly when distillation is disabled, establishing the no-regression contract.

1.3 What this paper is not

This is not a fine-tuning paper. The trainable parameter count is deliberately two orders of magnitude below full LoRA fine-tuning [2, 12] and the corpus size is two to three orders of magnitude below standard instruction-tuning runs. The target is the gap between calibration-free and calibration-aware compression at fixed runtime structure, not the gap between calibration-aware compression and full fine-tuning. The two questions have different answers and different costs.

2 Method

2.1 Phase 1: GRC projection (unchanged from Paper A)

For each attention block ℓ build the shared basis $\mathbf{P}_t^{(\ell)}$ from the joint Gram of $\mathbf{W}_Q^{(\ell)}, \mathbf{W}_K^{(\ell)}, \mathbf{W}_V^{(\ell)}$ and form the projected weights

$$\mathbf{W}_X'^{,(\ell)} = \mathbf{W}_X^{(\ell)} \mathbf{P}_k^{(\ell)} (\mathbf{P}_k^{(\ell)})^\top, \qquad X \in \{Q, K, V\}.$$

Optionally exempt the top-T highest-L2 columns from projection (sink-aware GRC, Paper A Table 6 / `sink_channel_pilot.json`). This phase is pure NumPy and runs on any host; it produces the same artefacts as Paper A and is included verbatim so that distillation can be disabled by a single command-line flag.

2.2 Phase 2: Teacher–student LoRA correction (new)

For each attention block ℓ and each slot $X \in \{Q, K, V\}$, fit adapters

$$\mathbf{A}_X^{(\ell)} \in \mathbb{R}^{m \times r}, \quad \mathbf{B}_X^{(\ell)} \in \mathbb{R}^{r \times d}$$

such that the merged weight

$$\hat{\mathbf{W}}_X^{(\ell)} \;=\; \mathbf{W}_X'^{,(\ell)} \;+\; \mathbf{A}_X^{(\ell)} \mathbf{B}_X^{(\ell)}$$

minimises a teacher–student logit MSE over a small calibration corpus. The teacher is the original uncompressed model; the student is the projected model with $\mathbf{W}_X'^{,(\ell)}$ frozen and only $(\mathbf{A}_X^{(\ell)}, \mathbf{B}_X^{(\ell)})$ trainable. Only the attention QKV adapters are fit; the output projection $\mathbf{W}_O$ and the FFN block remain at baseline, in keeping with Paper A's attention-only scope.

The default schedule is r=8, sequence length 512, batch 8, 500 AdamW steps with η=10^{-4}, weight decay 0, cosine LR with 100-step warmup, on a 5,000 to 10,000 token slice of the WikiText-2 train split [1]. The total trainable parameter count is

$$3 \cdot r \cdot d \cdot L \;=\; 3 \cdot 8 \cdot 4096 \cdot 32 \;\approx\; 3.1\,\mathrm{M}$$

for L=32 Llama-3.1-8B layers and r=8, two orders of magnitude below full LoRA fine-tuning and three

below full-rank fine-tuning. The training corpus is two to three orders of magnitude below standard instruction-tuning runs; this is intentional and is the basis of the "light" qualifier in the paper title.

2.3 Phase 3: Merge and ship

The trained LoRA factors are folded into the projected weights, producing a single matrix $\hat{\mathbf{W}}_X^{(\ell)}$ per slot per layer. The matrix is re-quantised to Q4_K_M via the standard `llama.cpp quantize` path and written into a fresh GGUF. The runtime loads the resulting GGUF unchanged; no runtime code change is required. If the user disables distillation the pipeline reduces to Paper A and produces the identical GGUF that Paper A produces, modulo quantisation determinism.

3 First-order bound on PPL gap closure

Let $\mathcal{L}_{\text{teach}}(x)$ and $\mathcal{L}_{\text{stud}}(x)$ be the teacher and the GRC-projected student log-likelihoods on a sequence x, and let $\Delta(x) = \mathcal{L}_{\text{teach}}(x) - \mathcal{L}_{\text{stud}}(x)$ be the per-sequence likelihood gap. The projection error decomposes layer-by-layer as

$$\mathbf{W}_X^{(\ell)} - \mathbf{W}_X'^{,(\ell)} \;=\; \mathbf{W}_X^{(\ell)}\big(\mathbf{I}_d - \mathbf{P}_k^{(\ell)}(\mathbf{P}_k^{(\ell)})^\top\big),$$

the Eckart–Young residual onto the orthogonal complement of the shared basis. By construction this residual has rank at most $d - k$ and Frobenius norm

$$\eta^{(\ell)} \;=\; \big\|\mathbf{W}_X^{(\ell)}(\mathbf{I}_d - \mathbf{P}_k^{(\ell)}(\mathbf{P}_k^{(\ell)})^\top)\big\|_F.$$

A rank-r LoRA adapter can absorb at most the leading r singular directions of this residual; the part of the residual that lies outside the top-r subspace is provably unrecoverable at fixed $(\hat{\mathbf{W}}_X)$ rank $k + r$. Writing $\sigma_i^{(\ell,X)}$ for the sorted singular values of the residual, the maximal achievable per-layer correction in Frobenius norm is

$$\eta_{\text{recov}}^{(\ell,X)} \;=\; \sqrt{\sum_{i=1}^{r}(\sigma_i^{(\ell,X)})^2} \;\le\; \eta^{(\ell,X)}.$$

Pulling this back through the (locally linearised) network gives a first-order upper bound on the achievable likelihood-gap closure:

$$\mathbb{E}_x[\Delta(x) - \Delta_{\text{distilled}}(x)] \;\le\; C \cdot \sum_{\ell,X} \eta_{\text{recov}}^{(\ell,X)},$$

where C is a model-dependent activation-Jacobian constant which we do not attempt to estimate analytically. The empirical content of the bound is the relative recoverable energy ratio

$$\rho \;=\; \frac{\sum_{\ell,X}\eta_{\text{recov}}^{(\ell,X)}}{\sum_{\ell,X}\eta^{(\ell,X)}},$$

which can be computed from the SVD of the projection residuals at zero runtime cost. For Llama-3.1-8B at k=1024, r=8, a numerical estimate (provided in the Phase 1 manifest) gives $\rho \approx 0.42$; that is, an r=8 LoRA can in principle absorb roughly 42% of the projection-discarded energy. The bound says nothing about whether the optimiser actually finds this optimum; it says only that the achievable PPL-gap closure is bounded above by what the residual energy permits.

We use ρ as a sanity check on the Phase 2 run: a run that closes much less than ρ is evidence that

the optimiser, the schedule, or the corpus is the bottleneck, not the projection. A run that closes much more than ρ would be evidence that the calibration corpus contains task-specific structure beyond what the WikiText-2 baseline measures (a useful failure mode to flag explicitly).

4 Why distillation does not break the fusion path

Paper A established that the GRC throughput improvement comes from two cooperating mechanisms: a partial L2-cache fit at small k, and kernel fusion of three Q/K/V `gemv_q4_k` launches into a fused `gemv_dual_q8_0` trio operating on k-dimensional intermediates. The fusion-viability condition is that the projected weights $\mathbf{W}'_X$ have rank k, with k satisfying the cache budget $S(k) = 2dk + 2k^2 \leq L_2$ on the target GPU.

The merged weights $\hat{\mathbf{W}}_X = \mathbf{W}'_X + \mathbf{A}_X\mathbf{B}_X$ in general have rank $k + r$ (when $r \leq m - k$). Three merge strategies are documented, and the runtime exposes a flag for each:

(a) Add-and-revert. Treat $\hat{\mathbf{W}}_X$ as a single full-rank matrix. The fusion path reverts to three independent `gemv_q4_k` launches, the cache-fit advantage is lost, and Paper A's +6.27% throughput gain reverts. This strategy exists only as a debugging baseline and is never the default.

(b) Fused-base plus side-LoRA. Keep $\mathbf{W}'_X$ on the fused path and run $\mathbf{A}_X\mathbf{B}_X$ as a separate small matmul-add. The side path's GEMV cost is bounded above by r/k of the main GEMV (since both consume the same activation, and the side path's output shape is r-then-d versus the main path's d-then-d). At r=8, k=1024 this is 0.78% worst-case relative overhead, well below the noise floor of Paper A's headline measurement. This is the default strategy.

(c) Re-quantised SVD-folded merge. Project $\hat{\mathbf{W}}_X$ back to rank k via a second SVD pass, accepting a merge-loss equal to the bottom r singular values of $\hat{\mathbf{W}}_X$. This strategy uses the lowest VRAM (no adapter storage) but partially defeats the distillation. It is reserved for the lowest-VRAM target and is not on the critical path of this paper.

The default merge strategy (b) means that the same model object can be loaded under either GRC alone or distilled GRC; the distilled correction is opt-in at runtime as well as at build time.

5 Falsifiable predictions

The Phase 2 run will produce numbers; the run is informative only if we specify in advance what numbers would refute the protocol. Three predictions are stated here; each can be falsified independently.

P1: Bound is tight to within an order of magnitude. We predict that the empirical PPL-gap closure at k=1024, r=8 on WikiText-2 will satisfy $0.1\rho \leq \text{closure}/\text{gap} \leq 1.0$, where ρ is the recoverable-energy ratio of § 3. Refutation: empirical closure outside this band. Two failure modes are informative: closure $< 0.1\rho$ flags optimiser or corpus pathology; closure > 1.0 flags task leakage from WikiText-2 calibration to WikiText-2 evaluation, in which case we will switch to a held-out calibration corpus.

P2: Throughput is preserved under merge strategy (b). We predict that distilled GRC at k=1024, r=8 measured under strategy (b) will satisfy $\text{thr}/\text{thr}_{\text{GRC}} \in [0.99, 1.01]$, that is, the side-LoRA overhead is below the noise floor of the n=8 paired bootstrap protocol. Refutation: a measured ratio outside

that band, in which case strategy (a) or (c) is preferred and the paper recommends a different default. The throughput baseline for this comparison is Paper A's GRC, not the uncompressed model.

P3: Sink-channel exemption and distillation are at least partly orthogonal. We predict that the PPL achieved by distilled GRC with sink-aware exemption (T=32, r=8, k=1024) is strictly lower than the PPL of either lever applied alone. Refutation: a measured PPL that is no better than the better of the two single-lever ablations. This would indicate that the LoRA fit is already absorbing the high-magnitude sink channels and that explicit exemption adds only redundancy.

A single Phase 2 run produces evidence on all three predictions simultaneously; the run is therefore worth the GPU budget regardless of which way the predictions break.

6 Empirical protocol

6.1 Done

- Phase 1 reference implementation. CPU-only NumPy runner in `scripts/grc_distill.py`; reproduces Paper A's GRC numbers when invoked with `-rank K` and no `-distill` flag. Expected wall clock: 60 to 120 s on a Ryzen 9 7940HS.
- Sink-channel pilot. Calibration-free top-T exemption, reported in Paper A Table 6 / `sink_channel_pilot.json`. Result: 1 to 3% relative reconstruction-error improvement at $k \in \{512, 1024, 1536\}$, T=32. The lever is small and orthogonal in this measure (see P3).
- Spectral diagnostics. The recoverable-energy ratio ρ defined in § 3, computed in `scripts/grc_distill.py -print-rho`, value $\rho \approx 0.42$ at k=1024, r=8.

6.2 Pending: Phase 2 PyTorch runner

Requires a GPU host with at least 24 GB VRAM to hold the teacher in BF16 with gradient activations on for the student. Reference targets: EC2 `g5.xlarge` (A10G 24 GB), `g6.xlarge` (L4 24 GB), or `p4d.24xlarge` (A100 40 GB). Estimated wall clock per run at the default schedule: 30 to 60 min on A100; 60 to 120 min on A10G.

6.3 Pending: Empirical numbers

The numbers below are predictions, not measurements. They will be replaced once the Phase 2 runner has executed; until then the abstract flags this section as scaffolded.

1. Vanilla GRC at k=1024: +13.30% PPL (Paper A baseline, measured).
2. Sink-aware GRC at k=1024, T=32: predicted +12.5% to +13.0% PPL.
3. Distilled GRC at k=1024, r=8, 5,000-token corpus: target +5% to +8% PPL (50 to 60% gap closure, in keeping with $\rho \approx 0.42$ and assuming the optimiser finds about half of the recoverable bound).
4. Distilled sink-aware GRC at k=1024, T=32, r=8: target +3% to +5% PPL.

6.4 Pending: Throughput re-measurement

Hypothesis: throughput is unchanged from vanilla GRC under merge strategy (b), per P2. The measurement will use the same n=8 paired bootstrap protocol as Paper A on the same RTX 4070 Laptop. If the measurement contradicts the hypothesis, that is the result.

7 Threat model and limitations

7.1 What the Phase 2 run will not establish

- Generalisation beyond WikiText-2. A single calibration corpus is not representative of all decoding workloads. The bound of § 3 is corpus-conditional; the recoverable energy ρ measured against WikiText-2 activations may differ from ρ on a code corpus, a chat corpus, or a mathematical-reasoning corpus. Generalisation across corpora is out of scope of the Phase 2 run and is flagged for follow-up.
- Generalisation beyond Llama-3.1-8B. The architecture assumptions (GQA [13], RoPE, d=4096, L=32) are baked into the trainable-parameter calculation. The cross-model SmolLM2 transfer check from Paper A applies to the projection step, not the distillation step.
- Generalisation beyond Q4_K_M. Re-quantisation determinism is assumed; the measured PPL after Phase 3 includes a re-quantisation noise term that we do not characterise.

7.2 Implementation hazards

- Teacher logits at FP16 are not bit-exact across hardware. The student trains against the teacher run on the exact same hardware (a single A10G or A100 instance). Cross-instance teacher caching is therefore unsafe and is disabled in the runner.
- The LoRA fit is sensitive to the AdamW β_2 default (0.999) at this corpus size; we use β_2=0.95 following QLoRA [12] guidance to suppress the rare large-update regime.
- Sink channels behave non-linearly under the LoRA fit if the fit is not warmstarted. We initialise $\mathbf{B}_X$ to zero (so the adapter is identity at step 0) and warmstart for 100 steps before turning on the LoRA path, following standard LoRA practice.

8 Reproducibility

The Phase 1 reference is in `scripts/grc_distill.py`. The Phase 2 runner will live in `scripts/grc_distill_runner` the EC2 driver in `scripts/ec2_paperE_distill/`. The companion reproduction guide on the project site walks through CPU-only and GPU paths step-by-step (see `https://nagusamecs.github.io/HyperTensor/research/repro/paper-e.html`). The pre-built W_proj caches from Paper A are published as GitHub Releases under the `wproj-cache-*` tag prefix and serve as the starting point for any Phase 2 run, eliminating the calibration-free projection step from the wall-clock budget.

A complete empirical reproduction will be published alongside the populated results once the runner has executed. The protocol, the predictions, and the bound in § 3 are checked in now, before the run, so that the run cannot retroactively become a post-hoc rationalisation.

References

[1] S. Merity, C. Xiong, J. Bradbury, and R. Socher. *Pointer Sentinel Mixture Models*. 2016. arXiv: 1609.07843 [cs.CL].

[2] E. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, and W. Chen. *LoRA: Low-Rank Adaptation of Large Language Models*. 2021. arXiv: 2106.09685 [cs.CL].

[3] M. Sun, X. Chen, J. Z. Kolter, and Z. Liu. *Massive Activations in Large Language Models*. 2024. arXiv: 2402.17762 [cs.CL].

[4] G. Xiao, Y. Tian, B. Chen, S. Han, and M. Lewis. *Efficient Streaming Language Models with Attention Sinks*. 2023. arXiv: 2309.17453 [cs.CL].

[5] E. Frantar, S. Ashkboos, T. Hoefler, and D. Alistarh. *GPTQ: Accurate Post-Training Quantization for Generative Pre-trained Transformers*. 2022. arXiv: 2210.17323 [cs.LG].

[6] J. Lin, J. Tang, H. Tang, S. Yang, X. Dang, and S. Han. *AWQ: Activation-Aware Weight Quantization for LLM Compression and Acceleration*. 2023. arXiv: 2306.00978 [cs.CL].

[7] G. Xiao, J. Lin, M. Seznec, H. Wu, J. Demouth, and S. Han. *SmoothQuant: Accurate and Efficient Post-Training Quantization for Large Language Models*. 2022. arXiv: 2211.10438 [cs.CL].

[8] Z. Yuan, Y. Shang, Y. Song, Q. Wu, Y. Yan, and G. Sun. *ASVD: Activation-Aware Singular Value Decomposition for Compressing Large Language Models*. 2023. arXiv: 2312.05821 [cs.CL].

[9] S. Ashkboos, M. L. Croci, M. G. do Nascimento, T. Hoefler, and J. Hensman. "SliceGPT: Compress Large Language Models by Deleting Rows and Columns". In: *ICLR*. 2024.

[10] Y.-C. Hsu, T. Hua, S. Chang, Q. Lou, Y. Shen, and H. Jin. "Language Model Compression with Weighted Low-Rank Factorization". In: *ICLR*. 2022.

[11] E. Frantar and D. Alistarh. *SparseGPT: Massive Language Models Can Be Accurately Pruned in One Shot*. 2023. arXiv: 2301.00774 [cs.LG].

[12] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer. *QLoRA: Efficient Finetuning of Quantized LLMs*. 2023. arXiv: 2305.14314 [cs.LG].

[13] J. Ainslie, J. Lee-Thorp, M. de Jong, Y. Zemlyanskiy, F. Lebrón, and S. Sanghai. *GQA: Training Generalized Multi-Query Transformer Models from Multi-Head Checkpoints*. 2023. arXiv: 2305.13245 [cs.CL].